

LLM-Based Multi-Agent Systems are Scalable Graph Generative Models

LLM-Based Multi-Agent Systems are Scalable Graph Generative Models (2024) proposes a framework called **GraphAgent-Generator (GAG)**, which uses LLM-powered multi-agent simulations to generate large-scale, dynamic, text-attributed graphs. The central insight is that if LLM agents interact in a structured environment (e.g., an online forum or social platform), the resulting communication patterns naturally form evolving graphs — enabling the synthesis of realistic benchmark-quality datasets.

Generating Dynamic Graphs via LLM-Based Agent Simulation

The GAG framework simulates the **temporal evolution of social networks** by letting LLM agents:

- Post and reply to one another
- Form new connections (edges)
- Enter or leave the environment (nodes)
- Use language as part of interaction (text attribution)

Each round of simulation ($k = 1, 2, \dots, K$) updates the graph B_{kB_k} by:

- Sampling environment context (e.g., "Welcome Week at University Forum")
- Having agents post new messages based on goals and history
- Enabling others to respond (reply edges)
- Adding new nodes (users) and edges (replies)

This results in **text-attributed dynamic graphs** that resemble real-world social graphs.

LLM-Forum Dataset

Using GAG, the authors generate **LLM-Forum**, a Reddit-style synthetic forum dataset featuring:

- 5,000 agents
- 10,000 posts

- 20,000 reply edges
- Rich user profiles and multi-turn conversations

The resulting graph exhibits realistic properties such as:

- Power-law degree distribution
- High clustering
- Topic-driven subgraphs
- Coherent and contextually appropriate natural language

Evaluation and Downstream Use

The paper evaluates whether models trained on GAG-generated graphs can generalize to real-world data. They test graph representation learning methods (e.g., HeteroMLP, GLEM, GNNs) on tasks like node classification.

Key findings:

- Synthetic graphs generated by GAG enable **comparable model performance** to real benchmarks (e.g., Reddit, OGBN-Arxiv)
- Graphs are structurally and semantically rich enough for use in GNN training and evaluation
- GAG simulations scale to graphs with **thousands of nodes**, far beyond prior LLM-based multi-agent setups

Why It Matters for the Survey

GAG is one of the first systems to use LLMs for **zero-shot generation of dynamic graphs**, where:

- **Graph evolution** (nodes, edges) emerges from multi-round simulation
- **Text and structure co-evolve** through LLM-driven interactions
- The system supports **large-scale generation**, creating realistic and usable graph datasets

This paper expands the survey scope beyond encoding existing dynamic graphs — showing how LLMs can be used to **create them from scratch**, offering a novel direction for synthetic benchmark generation and agent-based graph simulation.